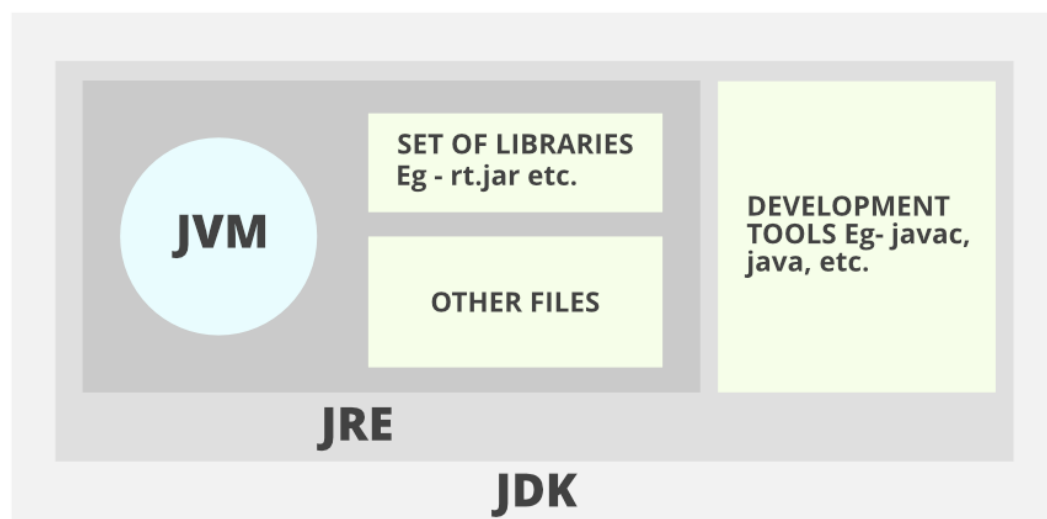


EXPERIMENT – 1

TITLE: Introduction to Java Environment

1. Explore and understand the role of JDK, JRE, and JVM.

- **JDK (Java Development Kit):** The JDK is a software development kit that provides tools for compiling, debugging, and running Java programs. It includes the JRE.
- **JRE (Java Runtime Environment):** The JRE is the environment required to run Java applications. It includes the JVM.
- **JVM (Java Virtual Machine):** The JVM is the runtime engine that executes Java bytecode. This is what makes Java platform-independent ("write once, run anywhere"). The JVM translates the bytecode into instructions that the underlying operating system can understand.



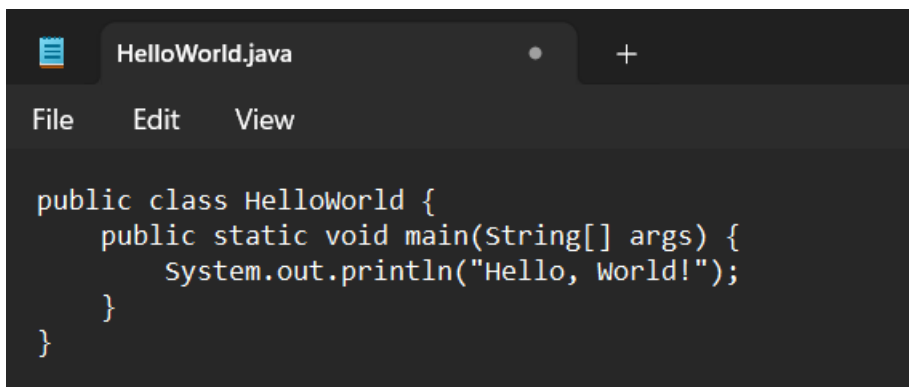
2. Install latest available JDK and verify the Java Environment.

1. **Download:** Download the latest JDK from Oracle's website (or a suitable alternative like Adoptium/Temurin).
2. **Installation:** Follow the installation instructions for your operating system (Windows, macOS, Linux).
3. **Verification:** Open a command prompt or terminal.
 - Type `javac -version` and press Enter. This should display the installed JDK version.

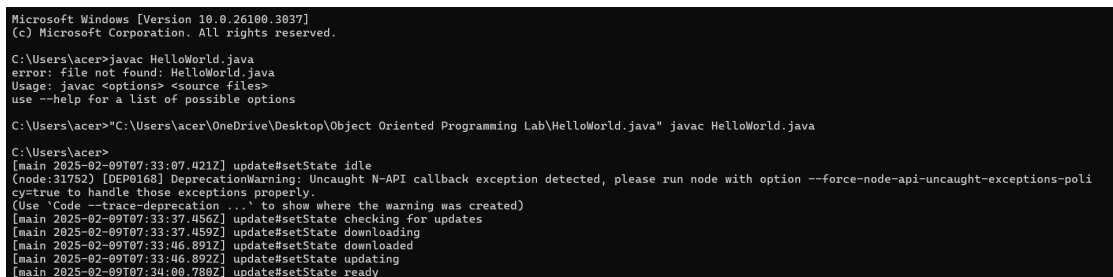
- Type `java -version` and press Enter. This should display the installed JRE version.

3. Create a Sample Hello World Program using simple text editor (e.g., Notepad) and show the steps to compile and execute the program using command prompt.

1. **Create the file:** Open a text editor (Notepad, VS Code, etc.) and type the following code:
2. **Save the file:** Save the file as `HelloWorld.java`.
3. **Compile:** Open a command prompt or terminal, navigate to the directory where you saved the file, and type `javac HelloWorld.java` and press Enter. This creates a `HelloWorld.class` file.
4. **Run:** In the same command prompt, type `java HelloWorld` and press Enter. This executes the program, and "Hello, World!" should be printed to the console.



```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```



```
Microsoft Windows [Version 10.0.26100.3037]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\acer>javac HelloWorld.java  
error: file not found: HelloWorld.java  
Usage: javac <options> <source files>  
use --help for a list of possible options  
  
C:\Users\acer>"C:\Users\acer\OneDrive\Desktop\Object Oriented Programming Lab\HelloWorld.java" javac HelloWorld.java  
  
C:\Users\acer>  
[main 2025-02-09T07:33:07.421Z] update#setState idle  
(node:31752) [DEP0168] DeprecationWarning: Uncaught N-API callback exception detected, please run node with option --force-node-api-uncaught-exceptions-policy=true to handle those exceptions properly.  
(Use 'Code --trace-deprecation ...' to show where the warning was created)  
[main 2025-02-09T07:33:37.456Z] update#setState checking for updates  
[main 2025-02-09T07:33:37.459Z] update#setState downloading  
[main 2025-02-09T07:33:46.891Z] update#setState downloaded  
[main 2025-02-09T07:33:46.892Z] update#setState updating  
[main 2025-02-09T07:34:00.788Z] update#setState ready
```

```

J HelloWorld.java 1 X
C: > Users > acer > OneDrive > Desktop > Object Oriented Programming Lab > J HelloWorld.java > ...
1  public class HelloWorld {
    Run | Debug
2      public static void main(String[] args) {
3          System.out.println(x:"Hello, World!");
4      }
5  }

```

```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SQL CONSOLE COMMENTS
PS C:\Users\acer> & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.14.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\acer\AppData\Local\Temp\vscodesws_6fca2\jdt_ws\jdt-
ls-java-project\bin' 'HelloWorld'
Hello, World!
PS C:\Users\acer>

```

4. Display your name and complete address in different lines.

Create a new Java file (e.g., MyInfo.java) with the following code (replace the bracketed information with your details):

Compile and run this program using the same steps as in step 3.

```

J Myinfo.java 1 X
C: > Users > acer > OneDrive > Desktop > Object Oriented Programming Lab > J Myinfo.java > Myinfo > main(String[])
1  public class Myinfo {
    Run | Debug
2      public static void main(String[] args) {
3          System.out.println(x:"Bhargav Das");
4          System.out.println(x:"Chaman Vihar, Phase 2");
5          System.out.println(x:"[Dehradun, Uttarakhand, 248001]");
6      }
7  }

```

```

PS C:\Users\acer> & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.14.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\acer\AppData\Local\Temp\vscodesws_4fca7\jdt-
ls-java-project\bin' 'Myinfo'
Bhargav Das
Chaman Vihar, Phase 2
[Dehradun, Uttarakhand, 248001]
PS C:\Users\acer>

```

